ASSIGNMENT-1

TOPIC: ASSESSING THE VULNERABILITY INDEX FOR THE SECC DATA OF RURAL INDIA.

COURSE NAME:

VULNERABILITY, ADAPTATION AND LIVELIHOODS

COURSE CODE: ECCSS 07

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1. INTRODUCTION

This report assesses the vulnerability index for the Socio-Economic Caste Census (SECC) data of rural India. The analysis follows the guidelines and methodologies outlined in the assignment brief, incorporating the Sudarshan and Iyengar weighting method and the IPCC vulnerability framework.

1.1. Data: https://secc.gov.in/getAllHhdSummaryNationalReport.htm

1.2. Assigned Tasks:

- ***i.*** Indicate the functional relationship between the indicators and vulnerability
- ***ii.*** Assign weight based on Sudarshan and Iyengar method.
- ***iii.*** Calculate the vulnerability using IPCC vulnerability framework.
- ***iv.*** Rank the states based on their vulnerability.
- ***v.*** Interpret the results based on the findings.

2. METHODOLOGICAL APPROACH

2.1. Identification of Functional Relationship

The relationship between different indicators and vulnerability is established based on their impact on social, economic, and environmental resilience. The selected indicators are categorized into exposure, sensitivity, and adaptive capacity.

Table 1. Functional Relationship Between Indicators and Vulnerability

S.NO.	INDICATORS	COMPONENT	FUNCTIONAL RELATIONSHIP	CODE
1	*Landless households deriving major part of their income from manual casual labour*	Sensitivity	Positive	S1
2	*Households with Destitutes/living on alms*	Sensitivity	Positive	S2
3	*Monthly income of highest earning household member is < 5000*	Sensitivity	Positive	S3
4	*Monthly income of highest earning household member is 5000 - 10000*	Sensitivity	Positive	S4
5	*Household without any phone*	Sensitivity	Positive	S5
6	*Total Households owning un-irrigated land (less reliable for agriculture)*	Sensitivity	Positive	S6
7	*Households with non-agricultural enterprises registered with government*	Adaptive Capacity	Negative	AC1
8	*Households paying income tax / professional tax*	Adaptive Capacity	Negative	AC2
9	*Households with salaried job in government*	Adaptive Capacity	Negative	AC3
10	*Households with salaried job in Public*	Adaptive Capacity	Negative	AC4

11	*Households with salaried job in Private*	Adaptive Capacity	Negative	AC5
12	*Monthly income of highest earning household member is > 10000*	Adaptive Capacity	Negative	AC6
13	*Household with government job and reporting monthly highest income > 5000*	Adaptive Capacity	Negative	AC7
14	*Households having motorized two/three/four wheelers and fishing boats*	Adaptive Capacity	Negative	AC8
15	*Households having 2-wheeler*	Adaptive Capacity	Negative	AC9
16	*Households having 3-wheeler*	Adaptive Capacity	Negative	AC10
17	*Households having 4-wheeler*	Adaptive Capacity	Negative	AC11
18	*Households having motorized fishing boat*	Adaptive Capacity	Negative	AC12
19	*Households that own a refrigerator*	Adaptive Capacity	Negative	AC13
20	*Households that own a landline phone*	Adaptive Capacity	Negative	AC14
21	*Households Owns mobile only*	Adaptive Capacity	Negative	AC15
22	*Households that own both landline and mobile*	Adaptive Capacity	Negative	AC16
23	*Households owning mechanized 3/4-wheeler agricultural equipment*	Adaptive Capacity	Negative	AC17
24	*Households having kisan credit card with the credit limit of Rs.50,000 and above*	Adaptive Capacity	Negative	AC18
25	*Total Households owning irrigated land*	Adaptive Capacity	Negative	AC19
26	*Total Households owning other land*	Adaptive Capacity	Negative	AC20
27	*Households with irrigation equipment*	Adaptive Capacity	Negative	AC21
28	*Household who has no land but have irrigation equipment*	Adaptive Capacity	Negative	AC22
29	*Household which doesn't own land but have Kisan Credit Card*	Adaptive Capacity	Negative	AC23

2.2. Normalisation of Data

The data available to us is largely heterogenous in nature and thus requires standardisation before analysis. This is done by Normalising the values of each Indicator for the respective states. According to the functional relationships, the following formulas are used for normalising:

i. Indicators having a positive functional relationship:

$$Normalized\ value\ (NV) = \frac{Actual\ value - Minimum\ value}{Maximum\ value - Minimum\ Value}$$

ii. Indicators having a negative functional relationship:

$$Normalized\ value\ (NV) = \frac{Maximum\ value - Actual\ value}{Maximum\ value - Minimum\ Value}$$

The normalised values are thus calculated for both the components, Sensitivity & Adaptive Capacity, separately. Please refer to "***Sheet 2- Normalisation of Sensitivity***" and "***Sheet 4- Normalisation of AC***" in the Excel Sheet.

2.3. Weight assignment using Sudarshan and Iyengar Method

The Sudarshan and Iyengar method is used to assign weights to different indicators based on their relative importance in assessing vulnerability. The weighting procedure involves normalization and aggregation.

In Iyengar and Sudarshan's method the weights are assumed to vary inversely as the variance over the regions in the respective indicators of vulnerability

Steps followed:

- Estimation of Standard Deviations (S.D) of the Normalised values for each Indicator.
- Taking the inverse of the standard deviations (1/S.D) followed by summation of each of the values [Σ(1/S.D)].
- The inverse of this summation is now calculated which is a Normalizing Constant (C).

$$Normalizing\ Constant\ (C) = \frac{1}{\Sigma(1/S.D)}$$

- The constant is divided by the Standard Deviations to the get the respective Weights.

$$Weights = \frac{C}{S.D}$$

- The Constant is the multiplied with each of the normalised values to get the weighted data.

Table 2. Weights for each indicator based on Sudarshan and Iyengar method

S.NO.	INDICATORS	CODE	WEIGHT
1	*Landless households deriving major income from manual casual labour*	S1	0.15202
2	*Households with Destitutes/living on alms*	S2	0.22604
3	*Monthly income of highest earning household member is < 5000*	S3	0.16285
4	*Monthly income of highest earning household member is 5000 - 10000*	S4	0.18249
5	*Household without any phone*	S5	0.14501
6	*Total Households owning un-irrigated land (less reliable for agriculture)*	S6	0.13157
7	*Households with non-agricultural enterprises registered with government*	AC1	0.04149
8	*Households paying income tax / professional tax*	AC2	0.03608
9	*Households with salaried job in government*	AC3	0.03932
10	*Households with salaried job in Public*	AC4	0.0452
11	*Households with salaried job in Private*	AC5	0.0373
12	*Monthly income of highest earning household member is > 10000*	AC6	0.0429
13	*Household with government job and reporting monthly highest income > 5000*	AC7	0.0394
14	*Households having motorized two/three/four wheelers and fishing boats*	AC8	0.0475
15	*Households having 2-wheeler*	AC9	0.0477
16	*Households having 3-wheeler*	AC10	0.0364
17	*Households having 4-wheeler*	AC11	0.0364
18	*Households having motorized fishing boat*	AC12	0.0401
19	*Households that own a refrigerator*	AC13	0.0369
20	*Households that own a landline phone*	AC14	0.0384
21	*Households Owns mobile only*	AC15	0.0461
22	*Households that own both landline and mobile*	AC16	0.0568
23	*Households owning mechanized 3/4-wheeler agricultural equipment*	AC17	0.0474
24	*Households having kisan credit card with the credit limit of ≥ Rs.50,000*	AC18	0.0515
25	*Total Households owning irrigated land*	AC19	0.0535
26	*Total Households owning other land*	AC20	0.0406
27	*Households with irrigation equipment*	AC21	0.05
28	*Household who has no land but have irrigation equipment*	AC22	0.0441
29	*Household which doesn't own land but have Kisan Credit Card*	AC23	0.0440

2.4. Calculation of Vulnerability Using the IPCC Framework

The vulnerability index is calculated as,

$$VI = [Sensitivity\ Index + Exposure\ Index] - Adaptive\ Capacity\ Index$$

Where,

i. ***Exposure*:** Represents the extent to which a system is subject to climate change impacts.

ii. ***Sensitivity*:** Determines how the system is affected by those impacts.

iii. ***Adaptive Capacity*:** Reflects the ability of the system to cope with or adapt to climate stressors.

Since, the data did not have any Exposure component, the formula is modified,

$$VI = Sensitivity\ Index + \frac{(1-Adaptive\ Capacity\ Index)}{2}$$

The weighted values of each components are then added for each state that gives the Index for that state.

3. RESULTS AND ANALYSIS

3.1. Sensitivity Ranking

Based on the computed Sensitivity Index, states are ranked to identify the most sensitive regions. This ranking further adds to the Inherent Vulnerability of each state.

Table 3. Sensitivity Indices (SI)

RANK	STATE NAME	SI
1	West Bengal	0.734532
2	Uttar Pradesh	0.705722
3	Bihar	0.594364
4	Maharashtra	0.53001
5	Madhya Pradesh	0.486614
6	Rajasthan	0.403174
7	Odisha	0.401378
8	Tamil Nadu	0.302408
9	Andhra Pradesh	0.298021
10	Karnataka	0.269283
11	Gujarat	0.255061
12	Assam	0.233956
13	Chhattisgarh	0.216495
14	Jharkhand	0.196718
15	Telangana	0.181137
16	Kerala	0.174276
17	Haryana	0.094972
18	Punjab	0.089251
19	Jammu & Kashmir	0.057979
20	Uttarakhand	0.047037
21	Himachal Pradesh	0.043608
22	NCT Of Delhi	0.027411
23	Tripura	0.021249
24	Meghalaya	0.016083
25	Manipur	0.011153
26	Nagaland	0.009506
27	Arunachal Pradesh	0.007209
28	Goa	0.004657
29	Puducherry	0.003048
30	Mizoram	0.002539
31	Sikkim	0.002187
32	Dadra & Nagar Haveli	0.001462
33	Andaman & Nicobar Islands	0.001362
34	Daman And Diu	0.000728
35	Chandigarh	0.000175
36	Lakshadweep	6.56E-05

3.2. Adaptive Capacity Ranking

Based on the computed Adaptive Capacity Index, states are ranked to identify the most resilient regions.

Table 4. Adaptive Capacity Indices (ACI)

RANK	STATE NAME	ACI
1	Chandigarh	0.999356
2	Lakshadweep	0.999232
3	Dadra & Nagar Haveli	0.997961
4	Daman and Diu	0.996894
5	Sikkim	0.996736
6	Mizoram	0.996462
7	Andaman & Nicobar Islands	0.995395
8	Puducherry	0.994313
9	Arunachal Pradesh	0.991803
10	Nagaland	0.988979
11	Meghalaya	0.987856
12	Manipur	0.983267
13	Goa	0.980835
14	Tripura	0.977709
15	Chhattisgarh	0.942333
16	Jammu & Kashmir	0.933366
17	Uttarakhand	0.932217
18	Himachal Pradesh	0.920813
19	NCT Of Delhi	0.865797
20	Odisha	0.862398
21	Haryana	0.843487
22	Assam	0.829706
23	Jharkhand	0.826485
24	Punjab	0.81047
25	Madhya Pradesh	0.757359
26	Telangana	0.754612
27	Andhra Pradesh	0.743524
28	Rajasthan	0.710922
29	Gujarat	0.707356
30	Karnataka	0.693698
31	West Bengal	0.683683
32	Tamil Nadu	0.675439
33	Kerala	0.635027
34	Bihar	0.5787
35	Maharashtra	0.539984
36	Uttar Pradesh	0.07581

3.3. Vulnerability Ranking

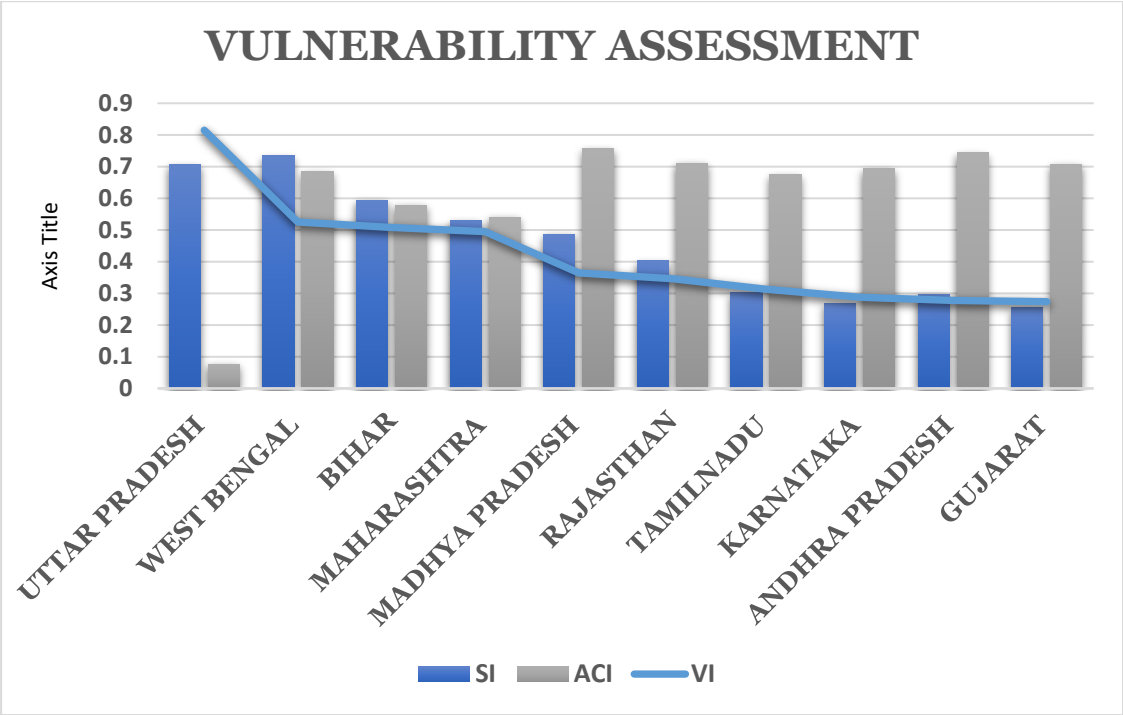
Based on the computed Vulnerability Index, states are ranked to identify the most vulnerable regions. This ranking helps policymakers focus on areas requiring urgent intervention.

Table 3. Vulnerability Indices (VI)

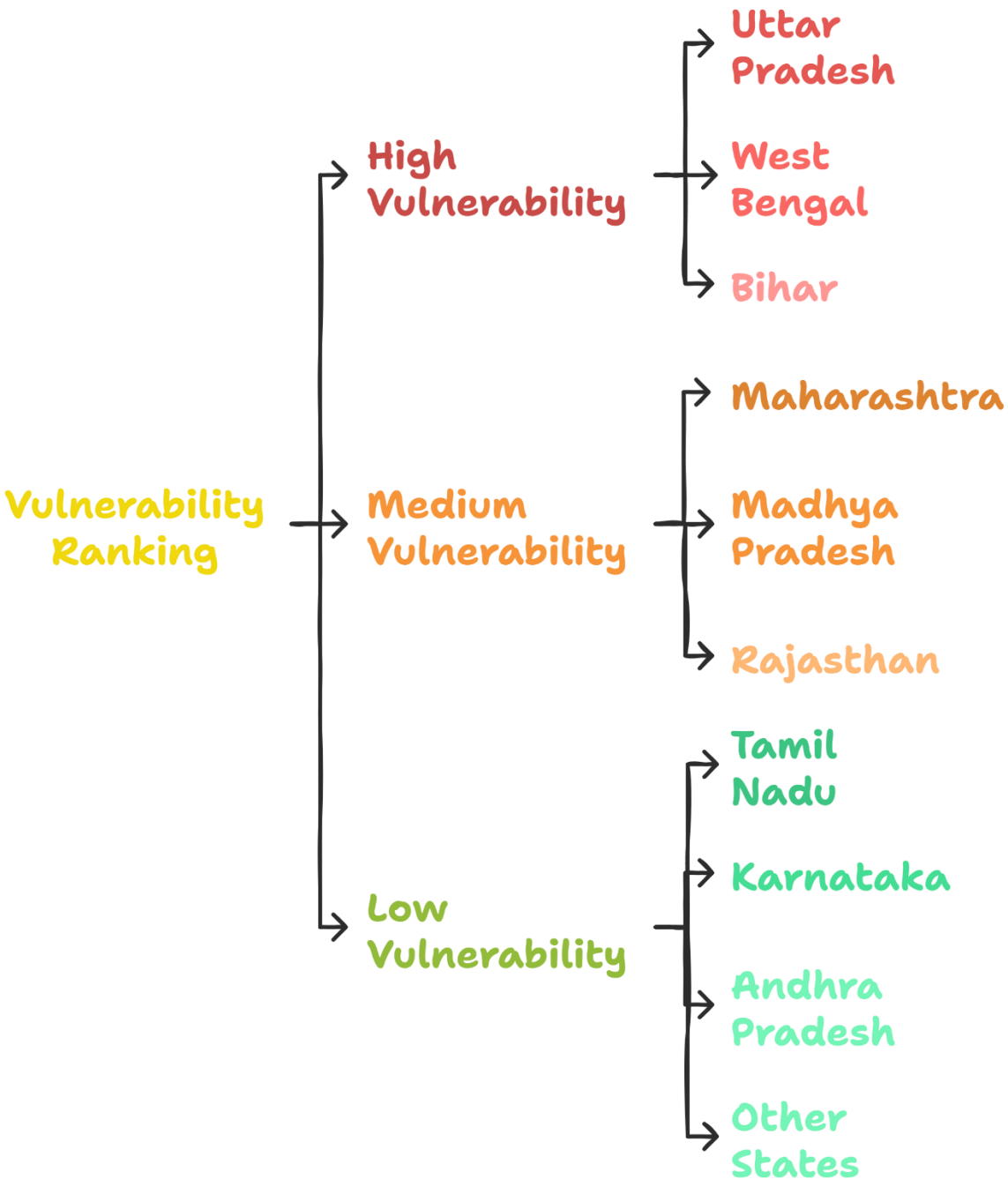
RANK	STATE NAME	VI
1	Uttar Pradesh	0.814956
2	West Bengal	0.525425
3	Bihar	0.507832
4	Maharashtra	0.495013
5	Madhya Pradesh	0.364627
6	Rajasthan	0.346126
7	Tamil Nadu	0.313484
8	Karnataka	0.287793
9	Andhra Pradesh	0.277248
10	Gujarat	0.273853
11	Kerala	0.269624
12	Odisha	0.26949
13	Telangana	0.213263
14	Assam	0.202125
15	Jharkhand	0.185116
16	Punjab	0.139391
17	Chhattisgarh	0.137081
18	Haryana	0.125743
19	NCT Of Delhi	0.080807
20	Jammu & Kashmir	0.062307
21	Himachal Pradesh	0.061397
22	Uttarakhand	0.05741
23	Tripura	0.02177
24	Meghalaya	0.014113
25	Manipur	0.013943
26	Goa	0.011911
27	Nagaland	0.010263
28	Arunachal Pradesh	0.007703
29	Puducherry	0.004368
30	Mizoram	0.003038
31	Andaman & Nicobar Islands	0.002983
32	Sikkim	0.002725
33	Daman and Diu	0.001917
34	Dadra & Nagar Haveli	0.00175
35	Lakshadweep	0.000417
36	Chandigarh	0.000409

4. DISCUSSION AND INSIGHTS

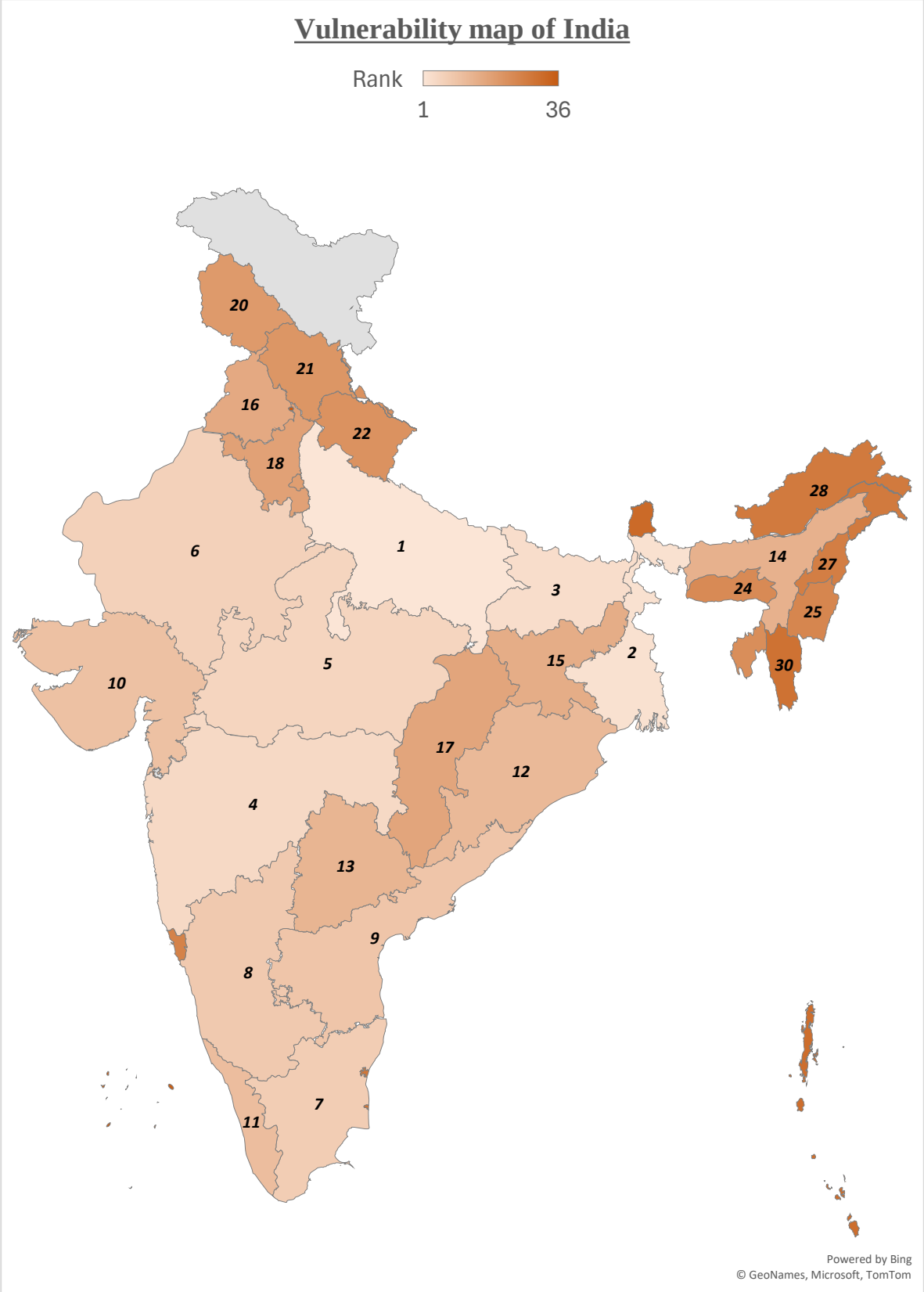
- The most vulnerable states exhibit common characteristics such as low Adaptive Capacities, implying low climate resilience, and high Sensitivities, implying high susceptibility to climate extremes.
- Certain states demonstrate resilience due to better infrastructure and social programs. Others show poor resilience due to low income, unemployment and poor infrastructural support.
- Although, pointing out relationships between each indicator and vulnerability followed by addressing each of them may be the obvious solution. Given the exclusive complex socio-economical systems existing in each of the state, this solution would prove to be ineffective as trade-offs would be required to be made between improving one indicator instead of another.
- This could lead to conflicts among policymakers and in-turn create communal unrest aggravated by long existing stereotypes regarding each state.
- Policy recommendations are thus required to address these conflicts of interests and disparities. State specific interventions that work in the bottom-top approach with efficient monitoring programs could be a potential solution to the problem highlighted.



Graph 1. Vulnerability assessment of the States



Flow Chart 1. Vulnerability ranking of Indian States



Map 1. Vulnerability map of India

5. CONCLUSION

This study aimed to assess the vulnerability index of rural India using SECC data through a structured approach, incorporating the Sudarshan and Iyengar weighting method and the IPCC vulnerability framework. The results revealed significant disparities in vulnerability across states, with Uttar Pradesh, West Bengal, and Bihar ranking among the most vulnerable due to their high sensitivity and low adaptive capacity. Conversely, states like Chandigarh and Lakshadweep demonstrated greater resilience due to better economic stability and infrastructural development.

The findings underscore the urgent need for targeted policy interventions to enhance adaptive capacity in highly vulnerable states. Addressing key indicators such as economic dependency, access to resources, and employment diversification can help mitigate vulnerability. Additionally, a **bottom-up policy approach**, considering regional socio-economic dynamics, is recommended to ensure effective and sustainable improvements in resilience.